

## PART II. SPECTRAL THEORY

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## CHAPTER I

## Preliminary Concepts

The study of linear operations requires a familiarity with certain basic concepts from the fields of set theory, topology and algebra. Chapter I discusses all of the concepts and results of these theories which will be used in this text. The exposition is complete, but brief; it contains results and proofs, with little accompanying illustrative or explanatory material. For the average reader, it will serve as a concise review of the topics treated, and, at the same time, as a collection of results available for easy reference. A reader familiar with set theory, metric spaces, and Hausdorff spaces, may prefer to start with Chapter II, and use Chapter I only for reference.

## A. SET-THEORETIC PRELIMINARIES

## 1. Notation and Elementary Notions

Rather than list the undefined terms of set theory, the axioms relating them, and the logical postulates governing the manipulation of these axioms, we shall in this first section follow an intuitive approach. The theorems and their proofs will be carefully stated although in an informal manner.

Upper or lower case Latin or Greek letters will usually be used to denote *sets*, *collections*, *families*, or *classes*, as well as to denote *functions* or *mappings*. The symbol  $\epsilon$  will indicate *membership* in a set; thus  $x \in A$  means that  $x$  is a member (or element) of the set  $A$ . If  $P(x)$  is a proposition concerning  $x$ , the symbol  $\{x|P(x)\}$  denotes the set of all  $x$  satisfying the proposition  $P(x)$ . The symbol  $\{x, y, \dots, z\}$  is used for the set whose elements are  $x, y, \dots, z$ . Where no danger of confusion exists, we sometimes write  $x$  in place of  $\{x\}$ . In this notation then  $\{x\} = \{y|y = x\}$ . The *void set* is the set with no members; it is denoted by  $\phi$ . If every element of a set  $A$  is also an element of a set  $B$ , then  $A$  is said to be *included* in  $B$ , or is said to be a *subset*

of  $B$ , and  $B$  is said to *contain*  $A$ . This is denoted symbolically by  $A \subseteq B$ , or  $B \supseteq A$ . Two sets are the same if and only if they have the same elements, i.e.,  $A = B$  if and only if  $A \subseteq B$  and  $B \subseteq A$ . The set  $A$  is said to be a *proper subset* of the set  $B$  if  $A \subseteq B$  and  $A \neq B$ . The notation  $A \subset B$ , or  $B \supset A$ , will mean that  $A$  is a proper subset of  $B$ . The *complement of a set  $A$  relative to a set  $B$*  is the set whose elements are in  $B$  but not in  $A$ , i.e., the set  $\{x | x \in B, x \notin A\}$ . This set is sometimes denoted by  $B - A$ . In a discussion where the set  $B$  is clearly understood, we may simply employ the phrase *complement of  $A$*  and use the symbol  $A'$  for this complement; this is stated symbolically  $A' = \{x | x \notin A\}$ . If  $A$  is a set whose elements are sets  $a$ , the set of all  $x$  such that  $x \in a$  for some  $a \in A$  is called the *union*, or *sum*, of the sets  $a$  in  $A$ . This *union* is denoted by  $\cup A$  or  $\bigcup_{a \in A} a$ . The *intersection*, or *product*, of the sets  $a$  in  $A$  is the set of all  $x$  in  $\cup A$  which are elements of every  $a \in A$ . If  $A = \{a, b, \dots, c\}$  we will sometimes write the union  $\cup A$  as  $a \cup b \cup \dots \cup c$ , and the intersection  $\cap A$  as  $a \cap b \cap \dots \cap c$ , or simply as  $ab \dots c$ . The operations of forming unions and intersections are *commutative* (i.e.,  $a \cup b = b \cup a$ ,  $ab = ba$ ), and *associative* (i.e.,  $a \cup (b \cup c) = (a \cup b) \cup c$ ,  $a(bc) = (ab)c$ ). Also, intersection is *distributive* with respect to union, and vice versa. This means that the following distributive laws hold:

$$x \cup \bigcap_{a \in A} a = \bigcap_{a \in A} (x \cup a), \quad x \cap \left( \bigcup_{a \in A} a \right) = \bigcup_{a \in A} (x \cap a).$$

Moreover, there are identities, known as the *rules of De Morgan*, which relate the operations of complementation, taking unions, and taking intersections. These rules are expressed by the formulas

$$\left( \bigcup_{a \in A} a \right)' = \bigcap_{a \in A} a', \quad \left( \bigcap_{a \in A} a \right)' = \bigcup_{a \in A} a',$$

where it is understood that all complements are taken relative to some set  $b$  which contains every element  $a$  of  $A$ .

Two sets are *disjoint* if their intersection is void. A family of sets is a *disjoint family* if every pair of distinct sets in the family is disjoint. A sequence  $\{a_n\}$  of sets is a *sequence of disjoint sets* if  $a_n \cap a_m = \phi$  for  $n \neq m$ . A set  $a$  is said to *intersect* a set  $b$  if  $ab \neq \phi$ .

The terms *function*, *mapping*, *transformation*, and *correspondence* will be used synonymously. The symbols  $f: A \rightarrow B$  will mean that  $f$  is a function whose *domain* is  $A$ , and whose *range* is contained in  $B$ ;

that is, for every  $a \in A$ , the function  $f$  assigns an element  $f(a) \in B$ . If  $f: A \rightarrow B$  and  $g: B \rightarrow C$ , then the mapping  $gf: A \rightarrow C$  is defined by the equation  $(gf)(a) = g(f(a))$  for  $a \in A$ . If  $f: A \rightarrow B$  and  $C \subseteq A$ , the symbol  $f(C)$  is used for the set of all elements of the form  $f(c)$  where  $c \in C$ . If  $f: A \rightarrow B$  and  $D \subseteq B$  then  $f^{-1}(D)$  is defined as  $\{x | x \in A, f(x) \in D\}$ . The set  $f(C)$  is called the *image* of  $C$  and the set  $f^{-1}(D)$  is called the *inverse image* of  $D$ . If  $f: A \rightarrow A$  and  $C \subseteq A$ , then  $C$  is said to be *invariant under  $f$*  in case that  $f(C) \subseteq C$ . The function  $f$  is said to map  $A$  *onto*  $B$  if  $f(A) = B$  and *into*  $B$  if  $f(A) \subseteq B$ . The function  $f$  is said to be an *extension* of the function  $g$  and  $g$  a *restriction* of  $f$  if the domain of  $f$  contains the domain of  $g$ , and  $f(x) = g(x)$  for  $x$  in the domain of  $g$ . The restriction of a function  $f$  to a subset  $A$  of its domain is sometimes denoted by  $f|A$ . If  $f: A \rightarrow B$ , and for each  $b \in f(A)$  there is only one  $a \in A$  with  $f(a) = b$ ,  $f$  is said to have an *inverse* or to be *one-to-one*. The *inverse function* has domain  $f(A)$  and range  $A$ ; it is defined by the equation  $a = f^{-1}(b)$ . Thus the domain and range of  $f^{-1}$  are the range and domain, respectively, of  $f$ . The *characteristic function*  $\chi_E$  of a set  $E$  is the real function defined by the equations  $\chi_E(s) = 1$ ,  $s \in E$ , and  $\chi_E(s) = 0$ ,  $s \notin E$ .

Sometimes, when the range of a transformation  $f: A \rightarrow B$  is to be emphasized at the expense of  $f$  itself and its domain, we shall write  $f(a)$  as  $b_a$ . Then  $f(A)$  is said to be an *indexed set*, and  $A$  is said to be a *set of indices*. If  $B$  is a collection of sets, the union  $\cup f(A)$  will sometimes be written as  $\bigcup_{a \in A} b_a$ , and  $\cap f(A)$ , as  $\bigcap_{a \in A} b_a$ .

A *relation* in (or on) a set  $A$  is a collection  $r$  of ordered pairs  $[x, y]$  of elements of  $A$ . It is customary to write  $xry$  for  $[x, y] \in r$ . Other symbols for relations are  $=$ ,  $\leq$ ,  $\subset$ ,  $\subseteq$ ,  $\infty$ , and  $\equiv$ .

We presuppose a familiarity with the real and complex number systems. By the *extended real number system* we mean the real numbers with the symbols  $+\infty$  and  $-\infty$  adjoined; by the *extended complex number system* we mean the complex numbers with the single symbol  $\infty$  adjoined. If  $A$  is a set of real numbers, then the *supremum*, or the *least upper bound*, of  $A$  is the smallest real number  $b$  such that  $a \leq b$  for all  $a$  in  $A$ ; if no such number exists, we take  $+\infty$  as the supremum. In either case we write  $\sup A$  or  $\text{lub } A$  to denote the supremum. Similar definitions are given for the *infimum*, or the *greatest lower bound*, of a set  $A$  which we denote by the notation  $\inf A$  or  $\text{glb } A$ .

If  $\phi$  is the void subset of the real numbers, it is conventional to take  $-\infty = \sup \phi$ ,  $+\infty = \inf \phi$ . If  $A$  is an infinite set of real numbers, then the symbol  $\limsup A$  denotes the infimum of all numbers  $b$  with the property that only a finite set of numbers in  $A$  exceed  $b$ ; the definition of the symbol  $\liminf A$  is similar. In particular, if  $A$  is a sequence  $\{a_n\}$  then  $\limsup A$  and  $\liminf A$  are usually denoted by

$$\limsup_{n \rightarrow \infty} a_n, \quad \liminf_{n \rightarrow \infty} a_n,$$

respectively. If  $a$  and  $b$  are extended real numbers, then the symbol  $(a, b)$  denotes the *open interval* defined by  $\{x | a < x < b\}$ ; the symbol  $[a, b]$  denotes the *closed interval* defined by  $\{x | a \leq x \leq b\}$ ; and the *semi-open intervals*  $(a, b]$  and  $[a, b)$  are given by  $\{x | a < x \leq b\}$  and  $\{x | a \leq x < b\}$ , respectively. If  $f$  is a real function defined on an open interval containing zero then the equations

$$\limsup_{x \rightarrow 0} f(x) = \inf_{a > 0} \sup f(( - a, a)),$$

$$\liminf_{x \rightarrow 0} f(x) = \sup_{a > 0} \inf f(( - a, a)),$$

$$\limsup_{x \rightarrow 0^+} f(x) = \inf_{a > 0} \sup f((0, a)),$$

$$\liminf_{x \rightarrow 0^+} f(x) = \sup_{a > 0} \inf f((0, a)),$$

define the symbols on their left sides. Similar definitions hold for the symbols  $\limsup_{x \rightarrow 0^-} f(x)$ ,  $\liminf_{x \rightarrow 0^-} f(x)$ . Finally if  $z$  is a complex number and  $z = x + iy$ , where  $x$  and  $y$  are real, then  $x$  and  $y$  are called the *real part* and the *imaginary part* of  $z$  and are denoted by  $\Re(z)$  and  $\Im(z)$ , respectively.

## 2. Partially Ordered Systems

1 DEFINITION. A *partially ordered system*  $(E, \leq)$  (henceforth abbreviated p.o.s.) is a non-empty set  $E$ , together with a relation  $\leq$  on  $E$ , such that

- (a) if  $a \leq b$  and  $b \leq c$  then  $a \leq c$ ,
- (b)  $a \leq a$ .

The relation  $\leq$  is called an *order relation* in  $E$ . The notation  $y \geq x$  is sometimes used in place of  $x \leq y$ .

2 DEFINITION. A *totally ordered* subset  $F$  of a p.o.s.  $(E, \leq)$  is a subset of  $E$  such that for every pair  $x, y \in F$  either  $x \leq y$  or  $y \leq x$ .

3 DEFINITION. If  $F$  is a subset of a p.o.s.  $(E, \leq)$ , then an element  $x$  in  $E$  is said to be an *upper bound* for  $F$  if every  $f$  in  $F$  has the property  $f \leq x$ . An upper bound  $x$  for  $F$  is said to be a *least upper*

similar fashion, the terms *lower bound* and *greatest lower bound* may be defined. As in the case of real numbers we denote the least upper bound of  $F$  by  $\sup F$  or  $\text{lub } F$  and the greatest lower bound of  $F$  by  $\inf F$  or  $\text{glb } F$ .

4 DEFINITION. An element  $x$  in  $E$  is said to be *maximal* if  $x \leq y$  implies  $y \leq x$ .

The family  $\mathcal{A}$  of all subsets of a set  $X$  affords an illustration of these concepts. The inclusion relation  $\subseteq$  between the sets contained in  $X$  makes the pair  $(\mathcal{A}, \subseteq)$  a partially ordered system. An upper bound for a subfamily  $\mathcal{B} \subseteq \mathcal{A}$  is any set containing  $\cup \mathcal{B}$ , and  $\cup \mathcal{B}$  is the only least upper bound of  $\mathcal{B}$ . Similarly,  $\cap \mathcal{B}$  is the only greatest lower bound of  $\mathcal{B}$ . The only maximal element of  $\mathcal{A}$  is  $X$ . In general, when dealing with a family of subsets of a given set, it will be supposed that they are ordered by inclusion, unless some other ordering is explicitly defined.

The following proof contains the central argument of this section.

5 THEOREM. Let  $f : E \rightarrow E$  have the property that  $f(x) \geq x$  where  $(E, \leq)$  is a non-void partially ordered system with the additional properties:

- ( $\alpha$ ) If  $a \leq b$  and  $b \leq a$  then  $a = b$ ;
- ( $\beta$ ) every totally ordered subset of  $E$  has a least upper bound.

Then there is a  $w$  in  $E$  with  $w = f(w)$ .

PROOF. Let  $a$  be an element of  $E$  which will remain fixed throughout the proof. A subset  $B$  of  $E$  with the following three properties will be called *admissible*.

- (i)  $a \in B$ .
- (ii)  $f(B) \subseteq B$ .
- (iii) Every least upper bound of a totally ordered subset of  $B$  is in  $B$ .

There exists a set, namely  $E$  itself, which has these properties. Also an intersection of admissible sets is admissible. Hence the intersection  $\mathcal{A}$  of all such sets is a *minimal* admissible set. The set  $\{x | x \in E, x \geq a\}$  is an admissible subset of  $E$ , and thus

- (iv)  $a \leq x, x \in \mathcal{A}$ .

Now let  $P = \{x | x \in \mathcal{A}; y \in \mathcal{A}, y < x \text{ imply } f(y) \leq x\}$  where  $y < x$  means that  $y \leq x$  and  $y \neq x$ . It will be shown that

- (v)  $x \in P, z \in \mathcal{A}$  imply either  $z \leq x$  or  $z \geq f(x)$ .

Fix  $x$  in  $P$ , and define  $B$  to be the set of all  $z$  in  $\mathcal{A}$  such that either  $z \leq x$  or  $z \geq f(x)$ . The condition (iv) shows that  $B$  has the property (i). The set  $B$  has the property (ii). For if  $z \geq f(x)$ , then  $f(z) \geq$

since  $x \in P$ . The set  $B$  has the property (iii), for if  $u$  is a least upper bound for the totally ordered subset  $F$  of  $B$ , then either  $y \leq x$  for every  $y \in F$ , in which case  $u \leq x$ , or else  $y \geq f(x)$  for some  $y \in F$ , in which case  $u \geq f(x)$ . Thus  $B$  is an admissible subset of  $A$ , and therefore  $B = A$ , which proves the assertion (v).

It will next be shown that  $P$  is admissible. The condition (i) is vacuously satisfied by  $P$ . To prove that  $P$  has the property (ii), let  $x \in P$ . It will be shown that if  $z \in A$ , and  $z < f(x)$ , then  $f(z) \leq f(x)$ . From (v), either  $z \geq f(x)$ , or  $z \leq x$ , so that if  $z < f(x)$ , then  $z \leq x$ . Then, since  $x \in P$ , if  $z < x$ , then  $f(z) \leq x \leq f(x)$ , and if  $z = x$ , then  $f(z) = f(x)$ . To verify that  $P$  has the property (iii), let  $v$  be a least upper bound for the totally ordered set  $F \subseteq P$ . To show that  $v \in P$ , let  $z \in A$ ,  $z < v$ . From (v), it is seen that every  $x \in F$  satisfies either  $z \leq x$ , or  $x \leq f(x) \leq z$ . The second alternative cannot be valid for every  $x$  in  $F$ , for then  $v \leq z$ . Hence, for some  $x$  in  $F$ ,  $z \leq x$ . If  $z < x$  then  $f(z) \leq x \leq v$  by the definition of  $P$ . If  $z = x$ , then since  $z \neq v$ , there is a  $y$  in  $F$  with  $z < y$ , in which case  $f(z) \leq y \leq v$ . Thus, in both cases  $f(z) \leq v$ , which proves that  $v \in P$ , and that  $P$  has the property (iii). Thus  $P$  is an admissible subset of  $A$ , and so  $P = A$ . Therefore, by (v) it is seen that for any two elements  $x, z$  in  $A$ , either  $z \leq x$ , or  $z \geq f(x) \geq x$ , which shows that  $A$  is totally ordered. If  $w$  is a least upper bound for  $A$ , then, since  $f(w) \in A$ ,  $w \leq f(w) \leq w$ , and  $f(w) = w$ . Q.E.D.

6 THEOREM. (Hausdorff maximality theorem). Every partially ordered system contains a maximal totally ordered subsystem.

A more explicit statement of the theorem follows: Let the family  $\mathcal{C}$  of totally ordered subsets of a p.o.s.  $(E, \leq)$  be made into a p.o.s.  $(\mathcal{C}, \subseteq)$ , by using the relation of inclusion between the elements of  $\mathcal{C}$  (which are subsets of  $E$ ). Then  $\mathcal{C}$  has a maximal element.

PROOF. If  $\mathcal{C}$  has no maximal element, then to every  $A \in \mathcal{C}$  there corresponds an  $f(A) \in \mathcal{C}$  containing  $A$  properly. Thus Theorem 5 is contradicted by the function  $f: \mathcal{C} \rightarrow \mathcal{C}$ . Q.E.D.

7 THEOREM. (Zorn's lemma). A partially ordered system has a maximal element if every totally ordered subsystem has an upper bound.

PROOF. Applying Theorem 6, let  $x$  be an upper bound of a maximal totally ordered subset  $E_0$  of the p.o.s.  $(E, \leq)$ . Let  $x \leq y$ . Then,

if  $y \notin E_0$ , the set  $E_0 \cup \{y\}$  is a totally ordered set containing  $E_0$  as a proper subset. Hence  $y \in E_0$ , so that  $y \leq x$ . Q.E.D.

8 DEFINITION. A p.o.s.  $(E, \leq)$  is said to be *well-ordered* if

(i)  $a \leq b$  and  $b \leq a$  implies  $a = b$ .

(ii) Any non-void subset of  $E$  contains a lower bound for itself.

The positive integers in their usual order afford a familiar example of a well-ordered system.

9 THEOREM. (Well-ordering theorem of Zermelo) Every set may be well-ordered.

The theorem means that for every set  $E$  there is an order relation  $\leq$  in  $E$  such that the p.o.s.  $(E, \leq)$  is well-ordered.

PROOF. Consider the family  $\mathcal{C}$  of all well-ordered sets  $(E_0, \leq_0)$  such that  $E_0 \subseteq E$ . We define the ordering  $<$  in  $\mathcal{C}$  by placing  $(E_0, \leq_0) < (E_1, \leq_1)$  if and only if

(i)  $E_0 \subseteq E_1$ ,

(ii)  $x, y \in E_0$ ,  $x \leq_0 y$  imply  $x \leq_1 y$ ,

and

(iii)  $x \in E_0$ ,  $y \notin E_0$ ,  $y \in E_1$  imply  $x \leq_1 y$ .

Under this ordering every totally ordered subfamily  $\mathcal{C}_0$  of  $\mathcal{C}$  has an upper bound. Indeed, it will be seen that this upper bound may be defined as  $(\cup \mathcal{C}_0, \leq')$ , where  $x \leq' y$ , whenever  $x$  and  $y$  both belong to some subset  $E_0 \in \mathcal{C}_0$ , and  $x \leq_0 y$  in the ordering  $\leq_0$  of that  $E_0$ . It is clear that if  $(\cup \mathcal{C}_0, \leq')$  belongs to  $\mathcal{C}$  it is an upper bound for  $\mathcal{C}_0$ . It will now be shown that it is well-ordered and hence belongs to  $\mathcal{C}$ . The statement  $x \leq' x$  for  $x \in \cup \mathcal{C}_0$  is clear. If  $x \leq' y$  and  $y \leq' z$ , then  $x, y \in E_0 \in \mathcal{C}_0$ ,  $y, z \in E_1 \in \mathcal{C}_0$ ,  $x \leq_0 y$ , and  $y \leq_1 z$ . Since  $\mathcal{C}_0$  is totally ordered it may be supposed that  $(E_0, \leq_0) < (E_1, \leq_1)$ , and then it is seen that  $x \leq_1 z$ , and thus that  $x \leq' z$ . If  $x \leq' y$ , and  $y \leq' x$ , then  $x, y \in E_0$ , and  $x, y \in E_1$ , with  $x \leq_0 y$ , and  $y \leq_1 x$ . Then, supposing that  $(E_0, \leq_0) < (E_1, \leq_1)$ , it follows that  $x = y$ .

Now, let  $F \subseteq \cup \mathcal{C}_0$ , and let  $F$  be non-void. Then, for some  $E_0 \in \mathcal{C}_0$ ,  $F \cap E_0 \neq \emptyset$ . The p.o.s.  $(E_0, \leq_0)$  is well-ordered. Let  $x_0 \in F \cap E_0$  be a lower bound for  $F \cap E_0$ , under the ordering  $\leq_0$ . Then, if  $y \in F$ ,  $y \notin F \cap E_0$ , it follows that  $x_0, y \in E_1$ , where  $(E_0, \leq_0) < (E_1, \leq_1)$ , so that  $x_0 \leq_1 y$ . Thus,  $x_0$  is a lower bound for  $F$  under the ordering  $\leq'$ . Thus it has been shown that  $\mathcal{C}_0$  has an upper bound.

Hence, by Theorem 7, there is a maximal well-ordered subset  $E_0$  of  $E$ . Now  $E_0 = E$  for if  $x$  is in  $E$  but not in  $E_0$  the ordering  $\leq_0$  in  $E_0$  may be extended to the set  $E_0 \cup \{x\}$  by defining  $y \leq_0 x$  for  $y \in E_0$ . Q.E.D.

### 3. Exercises

1 If  $(E, \leq)$  is a p.o.s. with the property that every totally ordered set has a lower bound, then  $E$  contains a minimal element.

2 If a family  $\mathcal{C}$  of subsets of a set has the property that  $A \in \mathcal{C}$  if and only if every finite subset of  $A$  belongs to  $\mathcal{C}$ , then  $\mathcal{C}$  contains a maximal element.

3 Derive Theorem 6 from the statement of Exercise 2.

4 Show that any one of the following implies any other one: Theorem 6, Theorem 7, Theorem 9, and Exercise 2.

5 Prove that if  $A$  and  $B$  are sets, there exists either a one-to-one mapping of  $A$  into  $B$ , or a one-to-one mapping of  $B$  into  $A$ . This is known as the *comparability theorem for cardinals*.

6 Show that there exists a one-to-one correspondence between any infinite set  $A$  and the set of all pairs  $(a, n)$ , where  $a \in A$  and  $n$  is an integer.

7 Let  $R$  be the set of real numbers. A subset  $S \subseteq R$  is called a *Hamel basis* if every real number  $r$  can be uniquely represented as  $r = \sum_1^n \alpha_i s_i$ , where  $s_i \in S$  and  $\alpha_i$  is rational. (a) Show that there exists a Hamel basis. (b) Show that there exists a discontinuous real function of a real variable satisfying the identity  $f(x+y) = f(x) + f(y)$ .

8 A family  $\mathcal{M}$  of subsets of a set  $X$  is said to have property  $(\alpha)$  if  $(\alpha)$   $X$  is not the union of a finite number of subsets in  $\mathcal{M}$ .

Show that if  $\mathcal{M}$  has property  $(\alpha)$ , there exists a maximal family  $\mathcal{N}$  of subsets of  $X$  with property  $(\alpha)$  which contains  $\mathcal{M}$ . Also, show that any maximal  $\mathcal{N}$  has the property

( $\beta$ ) If  $A_i \subseteq X$ ,  $i = 1, \dots, n$ , and  $A_1 \cap \dots \cap A_n \in \mathcal{N}$ , then some  $A_i \in \mathcal{N}$ .

9 DEFINITION. A p.o.s.  $(E, \leq)$  is said to be *complete* if:

(i)  $a \leq b$  and  $b \leq a$  imply  $a = b$ ,

(ii) Every non-void subset has a least upper bound and a greatest lower bound.

10 (Tarski) If  $(E, \leq)$  is a complete p.o.s.,  $f: E \rightarrow E$ , and

$x \leq y$  implies  $f(x) \leq f(y)$ , then  $f$  has a fixed element  $x_0$  ( $f(x_0) = x_0$ ), and the set of all fixed elements contains its least upper bound and its greatest lower bound.

11 DEFINITION. For each  $x$  in a set  $X$  let  $A_x$  be a subset of a set  $A$ . Then the *Cartesian product*  $P_{x \in X} A_x$  or  $PA_x$  is defined to be the set of all functions  $f$  on  $X$  to  $A$  for which

$$f(x) \in A_x, \quad x \in X.$$

If  $X$  consists of a finite number of elements,  $X = \{x_1, \dots, x_n\}$ ,  $P A_x$  will sometimes be written  $A_{x_1} \times A_{x_2} \times \dots \times A_{x_n}$ .

12 If  $X$  is the set  $\{1, \dots, n\}$ , then  $P_{x \in X} A_x$  can be regarded as the set of  $n$ -tuples of the form  $[a_1, \dots, a_n]$ , where  $a_i \in A_i$ . If  $X$  is the set of integers, then  $PA_x$  may be regarded as the set of sequences  $[a_1, a_2, \dots]$ , where  $a_i \in A_i$ .

13 Let  $Q_x, N_x$ , and  $M_x$  be subsets of  $A_x$ , for  $x \in X$ . Let  $M = PM_x$ ,  $N = PN_x$ , and  $Q = PQ_x$ . Suppose that  $N \neq \phi \neq Q$ ,  $M = N \cup Q$ , and  $N \cap Q = \phi$ . Then there exists a uniquely determined  $x_0 \in X$ , such that

$$(a) \quad M_{x_0} = N_{x_0} \cup Q_{x_0}, \quad N_{x_0} \cap Q_{x_0} = \phi,$$

$$(b) \quad M_x = N_x = Q_x \text{ if } x \neq x_0, \quad x \in X.$$

14 DEFINITION. If  $Y \subseteq X$  the mapping which takes each  $f$  in  $P_{x \in X} A_x$  into its restriction  $f|Y$  is called the *projection* of  $P_{x \in X} A_x$  onto  $P_{x \in Y} A_x$ . This mapping is denoted by  $pr_Y$ . If  $Y = \{x\}$  then  $pr_Y$  will be written as  $pr_x$ .

15 Let  $X \neq \phi$ .

(a)  $PM_x$  is void if and only if some  $M_x$  is void.

(b) If  $M_x \subseteq N_x$  for  $x \in X$ ,  $P M_x \subseteq P N_x$ .

(c) If  $P M_x \neq \phi$ , the converse of (b) is true, and equality in the conclusion of (b) implies equality in the hypothesis.

(d) A set  $F$  is of the form  $P B_x$  with  $B_x \subseteq A_x$  if and only if  $F = P pr_x(F)$ .

## B. TOPOLOGICAL PRELIMINARIES

## 4. Definitions and Basic Properties

1 DEFINITION. A family  $\tau$  of subsets of a set  $X$  is called a *topology* in  $X$  if  $\tau$  contains the void set  $\phi$ , the set  $X$ , the union of every one of its subfamilies, and the intersection of every one of its finite subfamilies. The pair  $(X, \tau)$  is called a *topological space*; but sometimes if  $\tau$  is understood, we refer to  $X$  as a topological space. If  $\tau, \tau_1$  are two topologies in  $X$ ,  $\tau$  is said to be *stronger*, or *larger*, than  $\tau_1$  and  $\tau_1$  is said to be *weaker*, or *smaller*, than  $\tau$  if  $\tau \supseteq \tau_1$ . The sets in  $\tau$  are called the *open sets* of  $(X, \tau)$ . A *neighborhood of the point  $p$*  is an open set containing  $p$ . A *neighborhood of the set  $A$*  is an open set containing  $A$ . If  $A$  is a subset of  $X$ , then a point  $p$  is a *limit point*, or a *point of accumulation*, of  $A$  provided every neighborhood of  $p$  contains at least one point  $q \neq p$ , with  $q \in A$ . The *interior* of a set in  $X$  is the union of its open subsets; a point in the interior of a set is called an *interior point* of the set.

2 LEMMA. A set in a topological space is open if and only if it contains a neighborhood of each of its points.

This lemma, and a number of those to follow, are immediate consequences of the definitions involved. Where this is the case, the proofs will be omitted.

3 DEFINITION. A set is said to be *closed* if its complement is open.

4 LEMMA. The intersection of any family of closed sets is closed, the union of any finite family of closed sets is closed, and  $\phi$  and  $X$  are closed.

5 LEMMA. If  $\mathcal{F}$  is any family of subsets of  $X$  having the properties of Lemma 4, and  $\tau$  is the family of complements of members of  $\mathcal{F}$ , then  $(X, \tau)$  is a topological space, and  $\mathcal{F}$  is the family of closed sets of this topology.

6 DEFINITION. A family  $\beta$  of subsets of  $X$  is said to be a *base* for the topology  $\tau$  if  $\beta \subseteq \tau$  and if every set in  $\tau$  is the union of some subfamily of  $\beta$ . The family  $\beta$  is said to be a *subbase* for the topology  $\tau$  if  $\tau$  is the smallest topology containing  $\beta$ . If  $\beta$  is a collection of neighbor-

hoods of a set  $A \subseteq X$  and every neighborhood of  $A$  contains a set in  $\beta$ , then  $\beta$  is called a *fundamental family of neighborhoods for  $A$* .

For example, the usual *topology of the real line* is the topology on  $(-\infty, +\infty)$  which has as a base all open intervals  $(a, b)$ , where  $a$  and  $b$  are arbitrary real numbers. Another base for this topology is obtained by taking  $a$  and  $b$  to be rational numbers. A subbase for this topology is given by all infinite intervals  $(-\infty, a)$ ,  $(b, +\infty)$ , where  $a$  and  $b$  are either real or rational. The *topology of the extended real numbers* has as a base the sets  $[-\infty, a)$ ,  $(a, b)$ ,  $(b, +\infty]$ , where  $a$  and  $b$  are real numbers. The complex numbers and the extended complex numbers are treated similarly.

7 LEMMA. If  $\beta$  is a family of subsets of  $X$ , and if  $\tau$  is the family of all unions of subfamilies of  $\beta$ , then  $\tau$  is a topology if and only if

- (i) for every pair  $U, V \in \beta$  and  $x \in U \cap V$  there is a  $W \in \beta$ , such that  $x \in W \subseteq U \cap V$ , and
- (ii)  $X = \cup \beta$ .

8 LEMMA. A family  $\beta$  is a subbase for the topology  $\tau$  if and only if  $\beta \subseteq \tau$  and every open set is a union of finite intersections of sets in  $\beta$ .

9 DEFINITION. The *closure*  $\bar{A}$  of a set  $A$  is the intersection of all closed sets containing  $A$ . The set of points in  $\bar{A}$  and not in the interior of  $A$  is called the *boundary* of  $A$ .

10 LEMMA. The closure operation has the properties:

- (a)  $\overline{A \cup B} = \bar{A} \cup \bar{B}$ , (b)  $\bar{A} \supseteq A$ ,
- (c)  $\bar{\bar{A}} = \bar{A}$ , (d)  $\bar{A} = A$  if  $A = \phi$ ,
- (e)  $p \in \bar{A}$  if and only if every neighborhood  $N(p)$  of  $p$  intersects  $A$ .

PROOF. Statement (b) is self-evident. Since  $\bar{A}$  is closed, (Lemma 4), it follows that  $\bar{\bar{A}} = \bar{A}$ . The set  $\bar{A} \cup \bar{B}$  contains  $A$  and is closed. Thus  $\overline{A \cup B} \supseteq \bar{A}$ , and similarly  $\overline{A \cup B} \supseteq \bar{B}$ . Therefore  $\overline{A \cup B} \supseteq \bar{A} \cup \bar{B}$ . Conversely,  $\bar{A} \cup \bar{B}$  is closed (Lemma 4), so  $\bar{A} \cup \bar{B} \supseteq \overline{A \cup B}$ , which proves (a). Statement (d) means simply that  $\phi$  is closed. Statement (e) follows immediately from Definitions 3 and 9. Q.E.D.

11 LEMMA. Let  $\mathcal{A}$  be the family of all subsets of  $X$ , and let  $A \rightarrow \bar{A}$  be a mapping  $\mathcal{A} \rightarrow \mathcal{A}$ , which has the four properties (a), . . . , (d) of Lemma 10. Then the family  $\mathcal{F} = \{A | A = \bar{A}\}$  has the properties

listed in Lemma 5, so that the family of complements of elements of  $\mathcal{F}$  is a topology. The set  $\bar{A}$  is the closure of  $A$  in this topology.

PROOF. Statements 10(b), (d) show that  $\phi, X \in \mathcal{F}$ , and 10(a) shows that  $A \cup B \in \mathcal{F}$  if  $A, B \in \mathcal{F}$ . It follows that any finite union of sets in  $\mathcal{F}$  is in  $\mathcal{F}$ . From 10(a), it follows that  $A \subset B$  implies  $\bar{A} \subset \bar{B}$ , and hence if  $A_\alpha \in \mathcal{F}$ , it is seen that

$$\bigcap_\alpha \bar{A}_\alpha \subseteq \bar{A}_\alpha = A_\alpha.$$

Thus

$$\bigcap_\alpha A_\alpha \subseteq \overline{\bigcap_\alpha A_\alpha} \subseteq \bigcap_\alpha \bar{A}_\alpha = \bigcap_\alpha A_\alpha,$$

and the family  $\mathcal{F}$  has all of the properties listed in Lemma 5. From that lemma it is seen that the family of complements of elements of  $\mathcal{F}$  form a topology. It remains to be shown only that  $\bar{A}$  is the closure of  $A$  in this topology. Now if  $B$  is closed, and  $B \supseteq \bar{A}$ , then  $\bar{B} \supseteq \bar{A} = \bar{A}$ , which shows that  $\bar{A}$  is closed, and is, in fact, the smallest closed set containing  $A$ . Q.E.D.

12 DEFINITION. If  $Y \subseteq X$ , and  $\tau$  is a topology for  $X$ , then the topology

$$\tau_Y = \{A | A = B \cap Y, B \in \tau\}$$

is called the *natural relative topology* of  $Y$  generated by  $\tau$ . A subset of  $Y$  is said to be *relatively open* if it is in  $\tau_Y$ ; *relatively closed* if its complement relative to  $Y$  is relatively open. Other terms like *relative closure* of a set are defined analogously. A topological space  $X$  is said to be *connected* if it is not the union of two non-void disjoint closed sets.

13 LEMMA. If  $Y \subseteq X$ , and  $(X, \tau)$  is a topological space, then the relative closure of a subset  $A$  of  $Y$  is the intersection of the closure of  $A$  with  $Y$ .

A subset of a topological space will always be regarded as a topological space with its relative topology, unless some other topology is given explicitly.

14 THEOREM. (Lindelöf) Let  $(X, \tau)$  be a topological space, and let  $\tau$  have a countable base  $\beta$ . Then every family  $\sigma \subseteq \tau$  contains a countable subfamily  $\sigma_0$  with  $\bigcup \sigma = \bigcup \sigma_0$ .

PROOF. Let  $B_1, B_2, \dots$  be an enumeration of  $\beta$ . Let  $\beta_0$  be the family of elements of  $\beta$  which are contained in some subset of  $\sigma$ . If  $B_n \in \beta_0$ , let  $C_n$  be some set in  $\sigma$  which contains  $B_n$ , and let  $\sigma_0$  be the family of all these  $C_n$ . Then it is evident that

$$\bigcup \sigma \supseteq \bigcup \sigma_0 \supseteq \bigcup \beta_0.$$

Since  $\beta$  is a basis, if  $p \in A \in \sigma$ , there is a  $B_n \in \beta$  such that  $p \in B_n \subseteq A$ , and so  $\bigcup \beta_0 \supseteq \bigcup \sigma$ . Q.E.D.

15 DEFINITION. If  $(X, \tau)$  and  $(Y, \tau_1)$  are topological spaces, and  $f: X \rightarrow Y$ , then  $f$  is *continuous* if  $f^{-1}(A) \in \tau$  for every  $A$  in  $\tau_1$ . In other words a mapping between topological spaces is continuous if the inverse image of every open set is open. The function  $f$  is said to be *continuous at the point  $x$*  if to every neighborhood  $U$  of  $f(x)$  there corresponds a neighborhood  $V$  of  $x$  with  $f(V) \subseteq U$ . If  $f$  is a continuous one-to-one map of  $X$  onto  $Y$ , and if the inverse function  $f^{-1}$  is also continuous, then  $f$  is called a *homeomorphism*, or a *topological isomorphism*. In this situation, the spaces  $X$  and  $Y$  are said to be *homeomorphic*.

16 LEMMA. Let  $X, Y$  be topological spaces and let  $f: X \rightarrow Y$ . Then, each of the following statements is equivalent to the continuity of  $f$ :

- (a) The function  $f$  is continuous at each point  $x \in X$ .
- (b) The inverse image of a closed set is closed.
- (c) If  $A \subseteq Y$ , then  $f^{-1}(\bar{A}) \supseteq \overline{f^{-1}(A)}$ .
- (d) If  $A \subseteq X$ , then  $f(\bar{A}) \subseteq \overline{f(A)}$ .
- (e) For every  $A$  in some subbase for the topology in  $Y$  the set  $f^{-1}(A)$  is open.

PROOF. If  $f$  is continuous and  $U$  is a neighborhood of  $f(x)$ , then  $V = f^{-1}(U)$  is a neighborhood of  $x$  with  $f(V) \subseteq U$ . Thus  $f$  is continuous at each point  $x$ . Conversely, Lemma 2 shows that (a) implies the continuity of  $f$ . Since the inverse image of a complement is the complement of the inverse image, the continuity of  $f$  is equivalent to the statement (b).

Statement (b) implies (c), since if  $f^{-1}(\bar{A})$  is closed, then  $\overline{f^{-1}(A)} \subseteq f^{-1}(\bar{A})$ . But (c) implies (b), since if  $A$  is closed,  $f^{-1}(A) \supseteq \overline{f^{-1}(A)}$ , so that  $f^{-1}(A)$  is closed.

Statements (c) and (d) are equivalent, for if  $f(\bar{A}) \subseteq \overline{f(A)}$ , then  $f(\overline{f^{-1}(B)}) \subseteq \bar{B}$ , and hence  $\overline{f^{-1}(B)} \subseteq f^{-1}(\bar{B})$ . On the other hand, if  $f^{-1}(\bar{A}) \supseteq \overline{f^{-1}(A)}$ , then  $f^{-1}(f(B)) \supseteq \overline{f^{-1}(f(B))} \supseteq \bar{B}$ , and hence  $\overline{f(B)} \supseteq f(\bar{B})$ .

It is clear that (a) implies (e). Also, since the inverse image of an intersection (or union) is the intersection (or union) of the inverse images, (e) implies (a). Q.E.D.

17 LEMMA. If  $X, Y, Z$  are topological spaces, and if  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are continuous, then the composite function  $fg$  is continuous.

The term *scalar* will be used for a real or complex number; a *scalar function* is a real or complex valued function. The topology in the set of scalars is always assumed to be that determined by the base whose elements are neighborhoods of the form  $\{\beta \mid |\beta - \alpha| < \epsilon\}$ .

18 LEMMA. Let  $f, g$  be continuous scalar functions on a topological space  $X$ , and let  $\alpha$  be a scalar. Then the functions given by the expressions

$$|f(x)|, \quad \alpha f(x), \quad f(x) + g(x),$$

are continuous. If  $f, g$  are real, then the expressions

$$\max(f(x), g(x)), \quad \min(f(x), g(x))$$

also define continuous functions.

## 5. Normal and Compact Spaces

Is there, on a given topological space, a continuous real function which is not a constant? If  $x$  and  $y$  are distinct points of a topological space  $X$ , is there a continuous real function on  $X$  with  $f(x) \neq f(y)$ ? If the answer to this second question is in the affirmative for an arbitrary pair of distinct points, it is said that there are enough real continuous functions to distinguish between the points of the space. It is not clear from any of the preceding remarks whether or not a given topological space has this property. However, it will be shown presently that the normal and the compact Hausdorff spaces introduced in this section all have enough continuous real functions to distinguish between their points.

1 DEFINITION. A topological space  $X$  is a *Hausdorff* space if it has the properties (a), (b), listed below. It is a *regular* space if it has the properties (a), (c), a *normal* space, if it has the properties (a), (d).

(a) Sets consisting of single points are closed.

(b) For every pair of distinct points  $x$  and  $y$ , there are disjoint neighborhoods of  $x$  and  $y$ .

(c) For every closed set  $A$ , and every  $x \notin A$ , there are disjoint neighborhoods of  $A$  and  $x$ .

(d) For every pair of disjoint closed sets  $A$  and  $B$ , there are disjoint neighborhoods of  $A$  and  $B$ .

2 THEOREM. (Urysohn) Let  $A$  and  $B$  be disjoint closed sets in a normal topological space  $X$ . Then there is a continuous real function  $f$  defined on  $X$ , such that  $0 \leq f(x) \leq 1$ ,  $f(A) = 0$ ,  $f(B) = 1$ .

PROOF. Let  $A_{1/2}$  and  $B_{1/2}$  be disjoint open sets, containing  $A$  and  $B$  respectively. Then we have

$$A \subseteq A_{1/2} \subseteq \bar{A}_{1/2} \subseteq B'_{1/2}, \quad B_{1/2} \supseteq B.$$

Then  $A$  and  $A'_{1/2}$  are disjoint closed sets, and  $B'_{1/2}$  and  $B$  are disjoint closed sets. Employing the hypothesis of normality again, we construct open sets  $A_{1/4}$ , and  $A_{3/4}$ , such that

$$A \subseteq A_{1/4} \subseteq \bar{A}_{1/4} \subseteq A_{1/2} \subseteq \bar{A}_{1/2} \subseteq A_{3/4} \subseteq \bar{A}_{3/4},$$

and  $\bar{A}_{3/4} \cap B = \phi$ . By induction, a family of open sets  $A_r$  may be defined for every dyadic rational  $r$ ,  $0 < r < 1$ , such that

$$(i) \quad r < s \text{ implies } \bar{A}_r \subseteq A_s$$

and

$$(ii) \quad A \subseteq A_r, \quad B \cap \bar{A}_r = \phi.$$

Let  $f(x) = 0$ , if  $x$  is in all of the sets  $A_r$ . Otherwise let

$$f(x) = \sup \{r \mid x \notin A_r\}.$$

To verify that  $f$  is continuous, let  $c = f(x)$  be positive. Then, for some suitable arbitrarily small  $\epsilon$  and  $\eta < \epsilon$ ,  $x$  is in the open set  $A_{c+\epsilon} \cap \bar{A}'_{c-\eta}$ . If  $y$  is also in this open set, then  $|f(x) - f(y)| < 2\epsilon$ . If  $f(x) = 0$  an analogous proof of continuity holds. Q.E.D.

3 THEOREM. (The Tietze extension theorem) If  $f$  is a bounded real continuous function defined on a closed set  $A$  of a normal space  $X$ , there

is a continuous real function  $F$  defined on  $X$  with  $F(x) = f(x)$  for  $x$  in  $A$ , and  $\sup_{x \in X} |F(x)| = \sup_{x \in A} |f(x)|$ .

PROOF. Since the theorem is obvious if  $f$  is identically zero on  $A$ , it will be assumed that this is not the case. Let  $f_0(x) = f(x)$ ,  $\mu_0 = \sup_{x \in A} |f_0(x)|$ ,  $A_0 = \{x | x \in A, f_0(x) \leq -\mu_0/3\}$ , and  $B_0 = \{x | x \in A, f_0(x) \geq \mu_0/3\}$ . Then  $A_0 \cap B_0 = \emptyset$ , and both are closed. Applying the previous theorem, we find a function  $F_0(x)$  defined on all of  $X$  such that  $F_0(A_0) = -\mu_0/3$ ,  $F_0(B_0) = \mu_0/3$ ,  $-\mu_0/3 \leq F_0(x) \leq \mu_0/3$ . Let  $f_1(x) = f_0(x) - F_0(x)$  for  $x \in A$ . Then  $f_1$  is continuous, and  $\mu_1 \leq (2/3)\mu_0$  where  $\mu_1 = \sup_{x \in A} |f_1(x)|$ .

By applying to the pair  $f_1, \mu_1$  the procedure applied to  $f_0, \mu_0$ , and then continuing inductively, one obtains a sequence  $F_i, i = 1, 2, \dots$ , of real continuous functions on  $X$ , with the properties:

$$|f(x) - \sum_{i=0}^n F_i(x)| \leq \left(\frac{2}{3}\right)^{n+1} \mu_0, \quad x \in A,$$

and

$$\sup_{x \in X} |F_n(x)| \leq \frac{1}{3} \left(\frac{2}{3}\right)^n \mu_0.$$

These properties show that the series  $\sum_{n=0}^{\infty} F_n(x)$  converges and defines a function  $F$  on  $X$  which coincides with  $f$  on  $A$ . To verify that  $F$  is continuous, let  $\varepsilon > 0$  and fix  $n$  so that  $2\mu_0(2/3)^{n+1} < \varepsilon/2$ . Then

$$\begin{aligned} |F(x) - F(y)| &\leq |F(x) - \sum_{i=0}^n F_i(x)| \\ &\quad + \left| \sum_{i=0}^n F_i(x) - \sum_{i=0}^n F_i(y) \right| + \left| \sum_{i=0}^n F_i(y) - F(y) \right| \\ &\leq 2\mu_0 \left(\frac{2}{3}\right)^{n+1} + \sum_{i=0}^n |F_i(x) - F_i(y)| \\ &< \frac{\varepsilon}{2} + \sum_{i=0}^n |F_i(x) - F_i(y)|. \end{aligned}$$

The continuity of  $F$  will be shown, using Lemma 4.16(a), by proving that  $F$  is continuous at each point  $x$ . According to Lemmas 4.16 and 4.18, there is a neighborhood  $V$  of  $x$  such that

$$\sum_{i=0}^n |F_i(x) - F_i(y)| < \frac{\varepsilon}{2}, \quad y \in V,$$

and thus  $|F(x) - F(y)| < \varepsilon$  for  $y$  in  $V$ . Q.E.D.

4 COROLLARY. A real continuous function defined on a closed subset of a normal space has a real continuous extension defined on the whole space.

PROOF. The only case where the theorem does not apply immediately is the case where  $f$  is unbounded. If  $f$  is real and continuous on the closed set  $A$  in the normal space  $X$ , the bounded function  $\arctan f(x)$  has a continuous extension  $\alpha(x)$  defined on  $X$ . The closed sets  $A$  and  $B = \{x | |\alpha(x)| = \pi/2\}$  are disjoint, and hence there is a continuous function  $\beta$  with  $0 \leq \beta(x) \leq 1$ , which vanishes on  $B$ , and has the constant value 1 on  $A$ . Thus the function  $\tan \beta(x)\alpha(x)$  is a continuous extension of  $f$ . Q.E.D.

5 DEFINITION. A covering of a set  $A$  in a topological space  $X$  is a family of open sets whose union contains  $A$ . The space  $X$  is said to be compact if every covering of  $X$  contains a finite subset which is also a covering. A topological space  $X$  is said to be locally compact if every point has a neighborhood whose closure is compact. A family of sets has the finite intersection property if every finite subfamily has a non-void intersection. A subset of  $X$  is called conditionally compact if its closure is compact in its relative topology. It should be noted that a subset  $A \subseteq X$  is compact in its relative topology if and only if every covering of  $A$  by open sets in  $X$  contains a finite subcovering.

A well-known example of a locally compact space is a closed set of real or complex numbers. Such a space is compact if and only if it is bounded. These statements constitute the Heine-Borel theorem.

6 LEMMA. A topological space is compact if and only if every family of closed sets with the finite intersection property has a non-void intersection.

This lemma follows immediately from the rules of DeMorgan, and the next lemma follows readily from the definitions involved.

- 7 LEMMA. (a) A closed subset of a compact space is compact.  
 (b) A continuous image of a compact space is compact.  
 (c) A compact subset of a Hausdorff space is closed.

8 LEMMA. *A continuous one-to-one map from a compact space to a Hausdorff space is a homeomorphism.*

PROOF. Let  $X$  be a compact space,  $Y$  a Hausdorff space, and  $f$  a one-to-one continuous function on  $X$ , with  $f(X) = Y$ . According to the preceding lemma, a closed set  $A$  in  $X$  is compact, its continuous image  $f(A)$  is compact, and, since  $Y$  is a Hausdorff space,  $f(A)$  is closed. Thus Lemma 4.16(b) shows that  $f^{-1}$  is continuous. Q.E.D.

9 THEOREM. *A compact Hausdorff space is normal.*

PROOF. Let  $A$  be a closed subset of the compact Hausdorff space  $X$ , and let  $p \notin A$ . Then, if  $q \in A$ , there is a neighborhood  $U_q$  of  $q$  and a neighborhood  $V_q$  of  $p$ , such that  $U_q \cap V_q = \emptyset$ . Since  $A$  is compact, a finite set  $U_{q_1}, \dots, U_{q_n}$  covers  $A$  and

$$(U_{q_1} \cup \dots \cup U_{q_n}) \cap (V_{q_1} \cap \dots \cap V_{q_n}) = \emptyset.$$

Thus any compact set  $A$  and any point  $p \notin A$  have disjoint neighborhoods. Let  $A$  and  $B$  be closed and disjoint. Then if  $p \in A$ , there is a neighborhood  $U_p$  of  $p$ , and a neighborhood  $V_p$  of  $B$ , such that  $U_p \cap V_p = \emptyset$ . Then a finite set  $U_{p_i}$  will cover  $A$ , and the sets  $U_{p_1} \cup U_{p_2} \cup \dots \cup U_{p_m}$  and  $V_{p_1} \cap V_{p_2} \cap \dots \cap V_{p_m}$  are disjoint neighborhoods of  $A$  and  $B$ , respectively. Q.E.D.

10 LEMMA. *A real continuous function on a compact space attains its supremum and its infimum.*

PROOF. Let  $f$  be a real continuous function on the compact space  $X$ . By Lemma 7(b) the set  $f(X)$  is compact and hence by Lemma 7(c) it is closed. Hence  $f(X)$  is a bounded closed set of real numbers, and, therefore, it contains both its supremum and its infimum. Q.E.D.

## 6. Metric Spaces

1 DEFINITION. Let  $X$  be a set, and let  $\rho$  be a real function on  $X \times X$ , with the properties:

- (i)  $\rho(x, y) \geq 0$ ,
- (ii)  $\rho(x, y) = 0$  if and only if  $x = y$ ,
- (iii)  $\rho(x, y) = \rho(y, x)$ , and
- (iv)  $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$ .

Then  $\rho$  is called a *metric*, or a *metric function* in  $X$ . Sets of the form

$$S(x, \varepsilon) = \{y | \rho(x, y) < \varepsilon\}$$

are called *spheres* in  $X$ . The sphere  $S(x, \varepsilon)$  has  $x$  for a *center*, and  $\varepsilon$  for a *radius*. The *metric topology* in  $X$  is the smallest topology containing the spheres. The set  $X$ , with its metric topology, is called a *metric space*. If  $X$  is a topological space such that there exists a metric function whose topology is the same as the original topology, we say that  $X$  is *metrizable*.

If  $A$  and  $B$  are subsets of a metric space, let  $\rho(A, B) = \inf_{a \in A, b \in B} \rho(a, b)$ .

If  $A$  is a subset of a metric space, the  $\varepsilon$ -neighborhood of  $A$  is the set  $S(A, \varepsilon) = \{x | \rho(A, x) < \varepsilon\}$ . The *diameter* of a set  $A$ , in symbols  $\delta(A)$ , is the number  $\sup_{a, b \in A} \rho(a, b)$ .

2 LEMMA. *If  $\rho$  is a metric function in the set  $X$ , the spheres form a base for the metric topology.*

PROOF. If  $u \in S(x, \varepsilon)S(y, \varepsilon')$ , choose  $\delta > 0$ , so that  $\rho(x, u) + \delta < \varepsilon$ ,  $\rho(y, u) + \delta < \varepsilon'$ . Then if  $v \in S(u, \delta)$ ,

$$\rho(x, v) \leq \rho(x, u) + \rho(u, v) < \rho(x, u) + \delta < \varepsilon,$$

and

$$\rho(y, v) \leq \rho(y, u) + \rho(u, v) < \rho(y, u) + \delta < \varepsilon',$$

which show that  $S(u, \delta) \subseteq S(x, \varepsilon)S(y, \varepsilon')$ . The desired conclusion follows from Lemma 4.7. Q.E.D.

3 THEOREM. *A metric space is normal.*

PROOF. If  $x \neq y$ ,  $S(x, (1/2)\rho(x, y))$  and  $S(y, (1/2)\rho(x, y))$  are disjoint neighborhoods of  $x$  and  $y$  respectively; therefore a metric space is a Hausdorff space. If  $A$  and  $B$  are disjoint closed sets, the sets  $A_1 = \{x | \rho(x, A) < \rho(x, B)\}$  and  $B_1 = \{x | \rho(x, B) < \rho(x, A)\}$ , are disjoint neighborhoods of  $A$  and  $B$ , respectively. Q.E.D.

4 LEMMA. *In a metric space, any subset, with its relative topology, is also a metric space.*

PROOF. The restriction of the metric function to a subset determines a metric, the topology of which is the relative topology of the subset. Q.E.D.

5 DEFINITION. A sequence  $\{a_n\}$  in a topological space is said to converge to a point  $a$  in the space if every neighborhood of  $a$  contains

all but a finite number of the points  $a_n$ . This notion is written symbolically  $a_n \rightarrow a$ , or  $\lim_{n \rightarrow \infty} a_n = a$ . A sequence  $\{a_n\}$  is said to be *convergent* if  $a_n \rightarrow a$  for some  $a$ . A sequence  $\{a_n\}$  in a metric space is a *Cauchy sequence* if  $\lim_{m, n} \varrho(a_m, a_n) = 0$ . If every Cauchy sequence is convergent, a metric space is said to be *complete*.

The next three lemmas are immediate consequences of the definitions.

6 LEMMA. In a metric space, a convergent sequence is a Cauchy sequence. A Cauchy sequence converges if and only if it has a convergent subsequence. A point  $a$  is in the closure of a set  $A$  in a metric space if and only if there is a sequence  $\{a_n\}$  of points of  $A$  converging to  $a$ .

7 LEMMA. A closed subspace of a complete metric space is complete. A complete subspace of a metric space is closed.

8 LEMMA. A mapping  $f: X \rightarrow Y$  between metric spaces is continuous at the point  $x$  if and only if  $\{f(x_n)\}$  converges to  $f(x)$ , whenever  $\{x_n\}$  converges to  $x$ .

9 THEOREM. (The Baire category theorem) If a complete metric space is the denumerable union of closed subsets, at least one of these closed subsets contains a non-void open set.

PROOF. Let  $X$  be a complete metric space with metric  $\varrho(x, y)$ . Let  $\{A_n\}$  be a sequence of closed sets whose union is  $X$ . For the purposes of an indirect proof, it is assumed that no  $A_n$  contains a non-void open set. Thus  $A_1 \neq X$ , and  $A_1'$  is open, and contains a sphere  $S_1 = S(p_1, \varepsilon_1)$  with  $0 < \varepsilon_1 < 1/2$ . By assumption, the set  $A_2$  does not contain the open set  $S(p_1, \varepsilon_1/2)$ ; hence the non-void open set  $A_2' \cap S(p_1, \varepsilon_1/2)$  contains a sphere  $S_2 = S(p_2, \varepsilon_2)$  with  $0 < \varepsilon_2 < 1/2^2$ . By induction, a sequence  $\{S_n\} = \{S(p_n, \varepsilon_n)\}$  of spheres is obtained with the properties

$$0 < \varepsilon_n < 1/2^n, \quad S_{n+1} \subseteq S(p_n, \varepsilon_n/2), \quad S_n A_n = \phi, \quad n = 1, 2, \dots$$

Since, for  $n < m$ ,

$$\begin{aligned} \varrho(p_n, p_m) &\leq \varrho(p_n, p_{n+1}) + \varrho(p_{n+1}, p_{n+2}) + \dots + \varrho(p_{m-1}, p_m) \\ &< \frac{1}{2^{n+1}} + \dots + \frac{1}{2^m} < \frac{1}{2^n}, \end{aligned}$$

The centers  $p_m$  form a Cauchy sequence, and hence converge to a point  $p$ . Since

$$\begin{aligned} \varrho(p_n, p) &\leq \varrho(p_n, p_m) + \varrho(p_m, p) \\ &< \frac{\varepsilon_n}{2} + \varrho(p_m, p) \rightarrow \frac{\varepsilon_n}{2} \end{aligned}$$

it is seen that  $p \in S_n$  for every  $n$ . This implies that  $p$  is in none of the sets  $A_n$ , and hence, that it is not in their union. This contradicts the assumption that  $X = \cup A_n$ . Q.E.D.

10 DEFINITION. A subset  $A$  of a topological space  $X$  is *sequentially compact*, if every sequence of points in  $A$  has a subsequence converging to a point of  $X$ .

11 DEFINITION. A set is said to be *dense* in a topological space  $X$ , if its closure is  $X$ . It is said to be *nowhere dense* if its closure does not contain any open set. A space is *separable*, if it contains a denumerable dense set.

12 THEOREM. If a topological space has a countable base, it is separable. Conversely, if a metric space is separable, it has a countable base. Thus a subspace of a separable metric space is separable.

PROOF. Let  $A_1, A_2, \dots$  be a base for the topology in a topological space  $X$ , and let  $p_n \in A_n$ . If  $V$  is open, there is a base element  $A_n$  contained in  $V$ , and hence a point  $p_n$  in  $V$ . Thus the denumerable set  $P = \{p_1, p_2, \dots\}$  is dense in  $X$ . Conversely, let  $p_1, p_2, \dots$  be a countable dense set in a metric space. It will be shown that the denumerable set of spheres  $S(p_n, r)$ , with  $r$  rational, forms a base. If  $p$  is in the open set  $V$ , then  $S(p, \varepsilon) \subseteq V$  for small  $\varepsilon$ . Some  $p_n$  is in  $S(p, \varepsilon/4)$ , and for such  $p_n$

$$p \in S(p_n, r) \subseteq S(p, \varepsilon) \subset V,$$

where  $r$  is any rational number between  $\varepsilon/4$  and  $\varepsilon/2$ . The final statement follows readily. Q.E.D.

13 THEOREM. A subset of a metric space is compact if and only if it is closed and sequentially compact.

PROOF. Let  $A$  be a compact subset of a metric space  $X$ . By Lemma 5.7(c),  $A$  is closed. If  $A$  is not sequentially compact, then some sequence  $\{a_n\}$  contains no convergent subsequence. Hence each point

in  $A$  has a neighborhood containing at most a finite number of  $a_n$ . Since a finite number of these neighborhoods cover  $A$ , the sequence  $\{a_n\}$  consists of only a finite number of distinct points of  $A$ , and therefore most certainly does have a convergent subsequence. This contradiction proves that  $A$  is sequentially compact.

Conversely, suppose that  $A$  is sequentially compact and closed. It will first be shown that  $A$  is separable. Let  $p_0$  be an arbitrary point of  $A$ , and let  $d_0 = \sup_{p \in A} \varrho(p_0, p)$ . The number  $d_0$  is finite, for if  $\varrho(p_0, q_n) \rightarrow \infty$ , there is a convergent subsequence  $q_{n_i} \rightarrow q$ , and hence  $\varrho(p_0, q) = \lim \varrho(p_0, q_{n_i}) = \infty$ , an impossibility.

Now, inductively, let  $p_{i+1}$  be chosen so that  $\min_{0 \leq n \leq i} \varrho(p_n, p_{i+1}) \geq d_i/2$ , where

$$d_i = \sup_{p \in A} \min_{0 \leq n \leq i} \varrho(p_n, p).$$

It is clear that  $d_0 \geq d_1 \geq \dots$ . If  $d_n \geq \varepsilon > 0$  for all  $n$ , then no subsequence of  $p_0, p_1, \dots$  is a Cauchy sequence; by Lemma 6 no subsequence converges. Since this contradicts our hypothesis, it follows that  $d_n \rightarrow 0$ . Thus, for every  $p$  in  $A$  and every  $\varepsilon > 0$ , there is a  $p_n$  such that  $\varrho(p_n, p) < \varepsilon$ . Therefore,  $\{p_0, p_1, \dots\}$  is a countable dense set.

Now, as is seen from Theorems 12 and 4.14, to prove that  $A$  is compact, it suffices to show that every countable covering of  $A$  by open sets  $G_1, G_2, \dots$  contains a finite subcovering. If  $\bigcup_{i=1}^n G_i \neq A$ , let  $x_n \notin \bigcup_{i=1}^n G_i$ ,  $x_n \in A$ . Let  $x_{n_i} \rightarrow x$  be a convergent subsequence of  $\{x_n\}$ . Then, since the complement of  $\bigcup_{i=1}^n G_i$  is closed,  $x \notin \bigcup_{i=1}^n G_i$ , so that  $x \notin \bigcup_{i=1}^\infty G_i$ ; this contradicts the fact that  $\bigcup_{i=1}^\infty G_i = A$ . Q.E.D.

14 DEFINITION. A subset  $K$  of a metric space is *totally bounded*, if for every  $\varepsilon > 0$  it is possible to cover  $K$  by a finite number of spheres  $S(k_i, \varepsilon)$ ,  $i = 1, \dots, n$ , with centers in  $K$ .

15 THEOREM. If  $K$  is a set in a metric space  $X$ , the following statements are equivalent:

- (a)  $K$  is sequentially compact:
- (b)  $\bar{K}$  is compact:
- (c)  $K$  is totally bounded and  $\bar{K}$  is complete.

Furthermore, a compact metric space is complete and separable.

PROOF. By the preceding theorem, (b) follows from (a) if we

can show that  $\bar{K}$  is sequentially compact. Let  $\{p_n\}$  be a sequence in  $\bar{K}$ , and let  $k_n \in K$  be such that  $\varrho(p_n, k_n) < 1/n$ ,  $n = 1, 2, \dots$ . Some subsequence of  $\{k_n\}$  converges, and it is readily seen that the same subsequence of  $\{p_n\}$  converges to the same point. This proves (b).

Suppose (b) is true and let  $\varepsilon > 0$ . Then, since  $K$  is dense in  $\bar{K}$ , the spheres  $S(k, \varepsilon)$ ,  $k \in K$ , cover  $\bar{K}$ . Consequently, a finite number  $S(k_1, \varepsilon), \dots, S(k_n, \varepsilon)$  of these spheres cover  $\bar{K}$  and hence  $K$ , proving that  $K$  is totally bounded. Let  $\{p_n\}$  be a Cauchy sequence in  $\bar{K}$ ; since  $\bar{K}$  is sequentially compact (Theorem 13), some subsequence of  $\{p_n\}$  converges to a point  $p_0 \in \bar{K}$ . It is readily seen that the entire sequence  $\{p_n\}$  converges to  $p_0$ , so  $\bar{K}$  is complete.

Let (c) be valid, and let  $\{k_n\}$  be a sequence in  $K$ . By the total boundedness of  $K$ , for each number  $1/n$ ,  $n = 1, 2, \dots$ , there is a covering of  $K$  by a finite collection of spheres with centers in  $K$  and radius  $1/n$ . Thus some subsequence  $\{k_{1,n}\}$  of  $\{k_n\}$  is contained entirely in a sphere of radius one; a subsequence  $\{k_{2,n}\}$  of  $\{k_{1,n}\}$  is contained entirely in a sphere of radius  $1/2$ , etc. Continue this process and set  $\bar{k}_n = k_{n,n}$ . By construction,  $\{\bar{k}_n\}$  is a Cauchy sequence; since  $\bar{K}$  is complete, it is convergent. Hence  $K$  is sequentially compact, and the equivalence of (a), (b) and (c) is proved.

To prove the final statement, observe that if a space  $X$  is compact, then from the equivalence of (b) and (c) it follows that  $X$  is complete. The fact that  $X$  is separable was proved in the course of the proof of Theorem 13. Q.E.D.

The selection procedure used to establish part (c) in the preceding proof is known as the *Cantor diagonal process*.

16 DEFINITION. A mapping  $f: X \rightarrow Y$  between metric spaces is said to be *uniformly continuous* on  $X$ , if to every  $\varepsilon > 0$  there corresponds a  $\delta > 0$  such that  $\varrho(x, x') < \delta$  implies  $\varrho(f(x), f(x')) < \varepsilon$ .

The next theorem is of an elementary character, and will be used frequently throughout the text.

17 THEOREM. (*Principle of extension by continuity*) Let  $X$  and  $Y$  be metric spaces, and let  $Y$  be complete. If  $f: A \rightarrow Y$  is uniformly continuous on the dense subset  $A$  of  $X$ , then  $f$  has a unique continuous extension  $g: X \rightarrow Y$ . This unique extension is uniformly continuous on  $X$ .

PROOF. If  $x \in X$ , there is a sequence of points  $a_n \in A$  with  $a_n \rightarrow x$ . Since  $\{a_n\}$  is a Cauchy sequence, and  $f$  is uniformly continuous, the sequence  $\{f(a_n)\}$  is also a Cauchy sequence. Since  $Y$  is complete, there is a point  $g(x) \in Y$  with  $f(a_n) \rightarrow g(x)$ . To verify that  $g(x)$  depends only upon  $x$ , and not upon the particular sequence  $a_n \rightarrow x$ , let  $\{b_n\}$  be another sequence in  $A$  with  $b_n \rightarrow x$ . Then  $\rho(a_n, b_n) \rightarrow 0$ , and, since  $f$  is uniformly continuous,  $\rho(f(a_n), f(b_n)) \rightarrow 0$ , and therefore  $f(b_n) \rightarrow g(x)$ . Now it is readily seen that  $\rho(x, x') < \delta$  implies  $\rho(g(x), g(x')) \leq \varepsilon$ ; from this the uniform continuity of  $g$  follows. Finally, the uniqueness of  $g$  is obvious. Q.E.D.

In general, a continuous function is not uniformly continuous, but on a compact metric space these two notions coincide.

18 THEOREM. If  $f: X \rightarrow Y$  is a continuous mapping between metric spaces, and if  $X$  is compact, then  $f$  is uniformly continuous.

PROOF. If  $f$  is not uniformly continuous, there exists an  $\varepsilon > 0$  and two sequences  $\{x_n\}$  and  $\{z_n\}$  such that  $\rho(x_n, z_n) < 1/n$  and  $\rho(f(x_n), f(z_n)) > \varepsilon$  for  $n = 1, 2, \dots$ . Since  $X$  is sequentially compact (Theorem 13), there exist subsequences  $\{x_{n_k}\}$  and  $\{z_{n_k}\}$  which converge, and it is evident that they converge to the same point. If  $f$  is continuous, then for sufficiently large  $k$ , we have  $\rho(f(x_{n_k}), f(z_{n_k})) < \varepsilon$  which contradicts the above inequality. Thus if  $f$  is not uniformly continuous, it is not continuous on  $X$ . Q.E.D.

The following result gives conditions under which the topology of a space can be given by a metric function.

19 THEOREM. (Urysohn metrization theorem) A regular topological space with a countable base is metrizable. In particular, a compact Hausdorff space is metrizable if and only if it has a countable base.

PROOF. We have seen in Theorem 15 that a compact metric space is separable and, by Theorem 12, has a countable base. On the other hand, a compact Hausdorff space is normal (5.9) and therefore regular (5.1), so the second statement follows from the first.

Let  $X$  be a regular space with a countable base  $\{U_n\}$ . It will first be shown that  $X$  is normal. Let  $A, B$  be closed and disjoint sets in  $X$ . Since  $X$  is regular there is for each point  $x$  in  $A$  a set  $U \in \{U_n\}$  with  $x \in U \subseteq \bar{U} \subseteq B'$ . Thus if  $\{V_n\}$  is the subsequence of  $\{U_n\}$  consisting of all  $U_n$  with  $\bar{U}_n \subseteq B'$  we have  $A \subseteq \bigcup_{n=1}^{\infty} V_n$ . Similarly

if  $\{W_n\}$  consists of those  $U_n$  with  $\bar{U}_n \subseteq A'$  then  $B \subseteq \bigcup_{n=1}^{\infty} W_n$ . Now let  $Y_1 = V_1$ ,  $Z_1 = W_1 - \bar{Y}_1$  and, inductively let  $Y_n = V_n - \bigcup_{j=1}^{n-1} \bar{Z}_j$ ,  $Z_n = W_n - \bigcup_{j=1}^n \bar{Y}_j$ . Thus  $Y = \bigcup_{n=1}^{\infty} Y_n$  and  $Z = \bigcup_{n=1}^{\infty} Z_n$  are open sets. It may be seen that  $Y$  and  $Z$  are disjoint by showing that  $Y_n Z_m = \phi$  for all  $n, m \geq 1$ . If  $n \leq m$  then  $Z_m \subseteq W_m - \bar{Y}_n \subseteq W_m - Y_n$  and so  $Y_n Z_m = \phi$ . If  $n > m$  then  $Y_n \subseteq V_n - Z_m \subseteq V_n - Z_m$  and so  $Y_n Z_m = \phi$ . To see that  $A \subseteq Y$  let  $x$  be a point of  $A$  and choose  $m$  with  $x$  in  $V_m$ . Then, since  $\bar{Z}_n \subseteq \bar{W}_n \subseteq A'$  for  $n \geq 1$ , and since  $x \in AV_m$ , we have  $x \in Y_m$  which proves that  $A \subseteq Y$ . Similarly  $B \subseteq Z$  and thus  $X$  is a normal space.

Again let  $U_1, U_2, \dots$  be a base for the open sets in  $X$ . If  $p \in U_m$ , then there exists a  $U_n$  such that  $p \in U_n \subseteq \bar{U}_n \subseteq U_m$ . Thus there exist pairs  $(U_n, U_m)$  of sets selected from the base with the property that  $\bar{U}_n \subseteq U_m$ ; but since there are only a countable number of sets in the base, there can only be a countable number of such pairs. Let

$$(U_{n_1}, U_{m_1}), \dots, (U_{n_k}, U_{m_k}), \dots$$

be an enumeration of such pairs, and for each  $k = 1, 2, \dots$ , by Theorem 5.2 there exists a continuous function  $f_k$  with  $f_k(\bar{U}_{n_k}) = 0$ ,  $f_k(U'_{m_k}) = 1$ ,  $0 \leq f_k(x) \leq 1$ . Let  $\rho$  be defined on  $X \times X$  by

$$\rho(x, y) = \sum_{k=1}^{\infty} 2^{-k} |f_k(x) - f_k(y)|, \quad x, y \in X.$$

It is evident that  $\rho$  satisfies (i), (iii) and (iv) of Definition 6.1. Suppose that  $\rho(x, y) = 0$  for some pair  $x \neq y$ ; then  $f_k(x) = f_k(y)$ ,  $k = 1, 2, \dots$ . On the other hand there exists a set  $U_{m(x)}$  from the base such that  $x \in U_{m(x)}$ ,  $y \notin U_{m(x)}$ . By the regularity of the space there is some other set  $U_{n(x)}$  from the base with  $x \in U_{n(x)} \subseteq \bar{U}_{n(x)} \subseteq U_{m(x)}$ , so that  $(U_{n(x)}, U_{m(x)})$  is one of the pairs listed above. Hence  $f_k(x) \neq f_k(y)$  for some choice of  $k$ , and this contradiction shows that  $\rho$  is a metric function on  $X$ .

Let  $x$  be given and  $\varepsilon > 0$ . Then it is readily seen that  $\rho(x, \cdot)$  is a continuous real valued function on  $X$ . Hence there exists a set  $U_m$  from the base with  $x \in U_m$  such that if  $y \in U_m$  then  $\rho(x, y) < \varepsilon$ . This shows that the identity mapping of  $X$  with the given topology onto  $X$  with the metric topology defined by  $\rho$  is a continuous function. On the other hand if  $x \in U_m$  there is a  $U_n$  such that  $x \in U_n \subseteq \bar{U}_n \subseteq U_m$ . Hence  $(U_n, U_m)$  occurs in the enumeration of the pairs, say in the

$k$ -th term. Then if  $\rho(x, y) < 2^{-k}$  we have  $|f_k(y)| < 1$  and so  $y \in U_m$ . Thus  $S(x, 2^{-k}) \subseteq U_m$ , which shows that the identity mapping of  $X$  with the metric topology onto  $X$  with the given topology is continuous. Hence the identity is a homeomorphism and the space is metrizable. Q.E.D.

## 7. Convergence and Uniform Convergence of Generalized Sequences

The notion of convergence introduced in Definition 6.5 is not sufficiently general for all our purposes. We wish to indicate various ways in which it can be generalized.

If  $X$  and  $Y$  are topological spaces and  $g : X \rightarrow Y$ , the expression  $\lim_{w \rightarrow x} g(w) = y$  is written to mean that for every neighborhood  $N_y$  of  $y$ , there exists a neighborhood  $N_x$  of  $x$ , such that  $g(N_x) \subseteq N_y$ . The following is a related, but more general notion: Let  $A$  be a set, and let  $X$  and  $Y$  be topological spaces. Let  $f : A \rightarrow X$ , and  $g : A \rightarrow Y$ . Then

$\lim_{f(a) \rightarrow x} g(a) = y$  is written to mean that for every neighborhood  $N_y$  of  $y$ , there exists a neighborhood  $N_x$  of  $x$ , such that  $g(f^{-1}(N_x)) \subseteq N_y$ . For instance, the statement  $\lim_{\rho(x, x_0) \rightarrow 0} x = x_0$  is true in every metric space. Of course, if  $A = X$ , and  $f$  is the identity mapping, then  $\lim_{f(a) \rightarrow x} g(a) = y$  if and only if  $\lim_{a \rightarrow x} g(a) = y$ .

The following definition gives a third important and interesting way in which the concept of convergence can be generalized:

1 **DEFINITION.** A partially ordered set  $(D, \leq)$  is said to be *directed*, if every finite subset of  $D$  has an upper bound. A map  $f : D \rightarrow X$  of a directed set  $D$  into a set  $X$  is called a *generalized sequence* of elements in  $X$ , or simply a generalized sequence in  $X$ . If  $f : D \rightarrow X$  is a generalized sequence in the topological space  $X$ , it is said to *converge to the point  $p$  in  $X$* , if to every neighborhood  $N$  of  $p$  there corresponds a  $d_0 \in D$ , such that  $d \geq d_0$  implies  $f(d) \in N$ . In this case, it is also said that the *limit of  $f$  exists and is equal to  $p$* , or, symbolically,  $\lim f(d) = p$ , or, if  $D$  is to be emphasized,  $\lim_D f(d) = p$ .

Each notion of convergence gives rise to a related notion of *uniform convergence*. For instance, let  $D$  be a directed set,  $A$  an arbitrary set, and  $X$  a metric space. Suppose that  $f = f(d, a)$  maps

$D \times A$  into  $X$ . Then the statement  $\lim_D f(d, a) = g(a)$  *uniformly on  $A$* , or uniformly for  $a \in A$ , means that for every  $\varepsilon > 0$  there exists a  $d_0 \in D$ , such that  $\rho(f(d, a), g(a)) < \varepsilon$  for  $d \geq d_0$  and for every  $a$  in  $A$ .

If  $f$  and  $g$  are two generalized sequences of real or complex numbers defined on the same directed set  $D$ , then we write

$$f = O(g)$$

in case there exists an  $A > 0$  such that  $|f(d)| \leq A|g(d)|$  whenever  $d \geq d_A$ . Similarly the symbol

$$f = o(g)$$

indicates that for every  $\varepsilon > 0$  there is a  $d_\varepsilon \in D$  such that  $|f(d)| \leq \varepsilon|g(d)|$  whenever  $d \geq d_\varepsilon$ .

In later chapters we will ordinarily denote a generalized sequence  $f : D \rightarrow X$  by the notation  $\{x_\alpha\}$ , thus emphasizing the range of the function.

Generalized sequences may be used in a general topological space in much the same way as ordinary sequences are used in a metric space. Several of the following lemmas illustrate this statement.

2 **LEMMA.** Let  $A$  be a set in a topological space. Then a point  $p$  is in the closure of  $A$  if and only if some generalized sequence in  $A$  converges to  $p$ .

**PROOF.** If  $p = \lim f(d)$ , with  $f(d)$  in  $A$ , then every neighborhood of  $p$  contains a point of  $A$ . Hence  $p \in \bar{A}$ . Conversely, let every neighborhood of  $p$  contain a point of  $A$ . Let the family  $\{N\}$  of neighborhoods of  $p$  be directed by defining  $N_1 \geq N_2$  to mean that  $N_1 \subseteq N_2$ . Let  $f$  be a function on  $\{N\}$ , whose value  $f(N)$  is a point in  $N \cap A$ . Then  $p = \lim f(N)$ . Q.E.D.

3 **LEMMA.** A topological space is a Hausdorff space if and only if no generalized sequence of its elements has more than one limit.

4 **LEMMA.** If  $X$  and  $Y$  are topological spaces, and  $f : X \rightarrow Y$ , then  $f$  is continuous if and only if  $\lim h(d) = x$  implies  $\lim f(h(d)) = f(x)$  for every generalized sequence  $h$  of elements of  $X$ .

The proofs of these lemmas are left to the reader.

Most of the notions related to the basic notion of convergence can be carried over from sequences to generalized sequences: Let

$f : D \rightarrow X$  be a generalized sequence of elements in a metric space  $X$ . We call  $f$  a *generalized Cauchy sequence* in  $X$ , if, for each  $\varepsilon > 0$ , there exists a  $d_0 \in D$ , such that  $\varrho(f(p), f(q)) < \varepsilon$  if  $p \geq d_0, q \geq d_0$ .

5 LEMMA. *If  $f$  is a generalized Cauchy sequence in a complete metric space  $X$ , there exists a  $p \in X$  such that  $\lim f(d) = p$ .*

PROOF. Let  $d_n \in D$  be such that  $c_1, c_2 \geq d_n$  implies  $\varrho(f(c_1), f(c_2)) < 1/n$ . Let  $b_n$  be an upper bound for the finite set  $\{d_1, d_2, \dots, d_n\}$ . Then it is evident that  $f(b_n)$  is a Cauchy sequence. Hence there exists a  $p \in X$  such that  $\lim_{n \rightarrow \infty} f(b_n) = p$ . Let  $\varepsilon > 0$  be given, and choose  $n_0$  such that  $2n_0^{-1} < \varepsilon$  and such that  $\varrho(f(b_{n_0}), p) < \varepsilon/2$ . Then if  $d \geq b_{n_0}$ , it is apparent that  $\varrho(f(d), p) < \varepsilon$ . Q.E.D.

The reader who reviews the preceding discussion of metric spaces will discover other results which can be carried over from sequences to generalized sequences.

The following important result on interchange of limit operations is due to E. H. Moore.

6 LEMMA. *Let  $D_1$  and  $D_2$  be directed sets, and suppose that  $D_1 \times D_2$  is directed by the relation  $(d_1, d_2) \leq (d'_1, d'_2)$ , which is defined to mean that  $d_1 \leq d'_1$  and  $d_2 \leq d'_2$ . Let  $f : D_1 \times D_2 \rightarrow X$  be a generalized sequence in the complete metric space  $X$ . Suppose that:*

- (a) *for each  $d_2 \in D_2$ , the limit  $g(d_2) = \lim_{D_1} f(d_1, d_2)$  exists, and*
- (b) *the limit  $h(d_1) = \lim_{D_2} f(d_1, d_2)$  exists uniformly on  $D_1$ .*

*Then the three limits*

$$\lim_{D_2} g(d_2), \quad \lim_{D_1} h(d_1), \quad \lim_{D_1 \times D_2} f(d_1, d_2)$$

*all exist and are equal.*

PROOF. Let  $\varepsilon > 0$  be given. There is a  $\delta_2 \in D_2$  such that  $d_2 \geq \delta_2$  implies  $\varrho(f(d_1, d_2), h(d_1)) < \varepsilon/8$  for all  $d_1 \in D_1$ . It follows that

$$\varrho(f(d_1, d_2), f(d_1, \delta_2)) < \varepsilon/4$$

for  $d_1 \in D_1$ . Thus if  $\delta_1 \in D_1$  is such that  $d_1 \geq \delta_1$  implies  $\varrho(f(d_1, \delta_2), g(\delta_2)) < \varepsilon/8$ , we have

$$\varrho(f(d_1, \delta_2), f(\delta_1, \delta_2)) < \varepsilon/4,$$

so that  $\varrho(f(d_1, d_2), f(\delta_1, \delta_2)) < \varepsilon/2$ . Hence, if  $d_1, d'_1 \geq \delta_1$  and  $d_2, d'_2 \geq \delta_2$ , we have  $\varrho(f(d_1, d_2), f(d'_1, d'_2)) < \varepsilon$ . Thus,  $f$  is a generalized Cauchy

sequence. By Lemma 5,  $\lim_{D_1 \times D_2} f(d_1, d_2) = p$  exists. We have

$$\varrho(p, f(d'_1, d'_2)) = \varrho(\lim_{D_1 \times D_2} f(d_1, d_2), f(d'_1, d'_2)) \leq \varepsilon, \quad d'_1 \geq \delta_1, \quad d'_2 \geq \delta_2.$$

Hence

$$\varrho(p, g(d'_2)) = \varrho(p, \lim_{D_1} f(d'_1, d'_2)) \leq \varepsilon, \quad d'_2 \geq \delta_2,$$

so that  $\lim_{D_2} g(d_2) = p$ .

In the same way, we show that  $\lim_{D_1} h(d_1) = p$ . Q.E.D.

7 COROLLARY. *Let  $D$  be a directed set, let  $X$  and  $Y$  be topological spaces, and let  $Y$  be complete and metric. Let  $f : D \times X \rightarrow Y$ , so that  $f(d, x)$  is a generalized sequence of functions on  $X$ , with values in  $Y$ . Suppose that:*

- (a) *for each  $d_0 \in D$ , the function  $f(d_0, x)$  is continuous, and*
- (b) *the limit  $g(x) = \lim_{D} f(d, x)$  exists uniformly on  $X$ .*

*Then  $g(x)$  is continuous.*  $D$

PROOF. This follows readily from Lemmas 4 and 6. Q.E.D.

It is often convenient to have a characterization of compactness in terms of generalized sequences. We will find it convenient to use one dealing with the notion of a cluster point.

8 DEFINITION. Let  $f : D \rightarrow X$  be a generalized sequence in a topological space  $X$ . A point  $p$  is a *cluster point* of  $f$  if for each neighborhood  $U$  of  $p$  and  $d_0 \in D$  there is a  $d \geq d_0$  such that  $f(d) \in U$ .

9 LEMMA. *A space  $X$  is compact if and only if each generalized sequence in  $X$  has a cluster point.*

PROOF. If  $f : D \rightarrow X$  is a generalized sequence in a compact space  $X$ , for each  $d \in D$  let

$$A_d = \{y | y = f(d'), d' \in D, d' \geq d\}.$$

The collection  $\{A_d\}$  and hence the collection  $\{\bar{A}_d\}$  have the finite intersection property. By Lemma 5.6 there is a point  $p$  common to their closures. Since every neighborhood of  $p$  intersects each  $A_d$ , the point  $p$  is a cluster point of  $f$ .

Conversely, let each generalized sequence in  $X$  have a cluster point. Let  $\mathcal{F}_1$  be a given family of closed sets with the finite intersection property, and let  $\mathcal{F}$  be the set of all finite intersections of

sets in  $\mathcal{F}_1$ . Then  $\mathcal{F}$  has the finite intersection property and is directed by inclusion. Picking a point in each set of  $\mathcal{F}$  we obtain a generalized sequence which therefore has a cluster point. This cluster point is clearly in each member of the family  $\mathcal{F}$ . Hence the sets in  $\mathcal{F}_1$  have a common point and  $X$  is compact. Q.E.D.

There is another notion which is often of convenience in discussions of convergence.

10 DEFINITION. A family  $\mathcal{C}$  of subsets of a set is said to be a *filter* if it possesses the following properties:

- (i) The void set  $\phi$  is not in  $\mathcal{C}$ ;
- (ii) if  $A \supseteq B$  and  $B \in \mathcal{C}$ , then  $A \in \mathcal{C}$ ;
- (iii) if  $A, B \in \mathcal{C}$ , then  $A \cap B \in \mathcal{C}$ .

If  $\mathcal{C}$  and  $\mathcal{C}'$  are filters of subsets of a set, we say that  $\mathcal{C}$  *refines*  $\mathcal{C}'$  if  $\mathcal{C} \supseteq \mathcal{C}'$ . A filter is called an *ultrafilter* if it is not refined by any filter but itself. A filter  $\mathcal{C}$  of subsets of a topological space  $X$  *converges to a point*  $p \in X$ , if every neighborhood of  $p$  belongs to  $\mathcal{C}$ .

The reader will observe that the collection of all subsets of a topological space  $X$  which contain a neighborhood of a point  $p$  forms a filter  $\mathcal{N}(p)$ . Thus the filter  $\mathcal{C}$  converges to  $p$  if and only if  $\mathcal{C}$  refines  $\mathcal{N}(p)$ , i.e.,  $\mathcal{C} \supseteq \mathcal{N}(p)$ . In applying the filter theory the following lemma, due to H. Cartan, is of considerable importance.

11 LEMMA. Every filter of subsets of a set  $X$  is refined by some ultrafilter of subsets of  $X$ .

PROOF. Let  $\mathcal{C}_0$  be a filter in  $X$ , and let  $\mathfrak{P}$  be the collection of all filters  $\mathcal{C}$  in  $X$  with  $\mathcal{C}_0 \subseteq \mathcal{C}$ . Let  $\mathfrak{P}$  be partially ordered by  $\subseteq$ . If  $\mathfrak{U} = \{\mathcal{C}\}$  is a totally ordered subset of  $\mathfrak{P}$ , then  $\mathcal{C}' = \{E \mid E \in \mathcal{C} \in \mathfrak{U}\}$  is readily seen to be a filter in  $\mathfrak{P}$  and  $\mathcal{C} \subseteq \mathcal{C}'$ , for all  $\mathcal{C} \in \mathfrak{U}$ . By Zorn's lemma (2.7) there is a maximal element which is obviously an ultrafilter in  $X$  and which refines  $\mathcal{C}_0$ . Q.E.D.

It is evident that if  $\mathcal{C}$  is an ultrafilter in  $X$  and if  $E \subseteq X$ , then precisely one of the sets  $E, E'$  is contained in  $\mathcal{C}$ .

12 LEMMA. A topological space  $X$  is compact if and only if every ultrafilter of subsets of  $X$  converges to a point in  $X$ .

PROOF. Let  $\mathcal{C}$  be an ultrafilter in a compact space  $X$ . If  $\mathcal{C}$  does not converge to  $p \in X$ , then some neighborhood  $N_p$  is not in  $\mathcal{C}$ , and thus its complement  $N'_p$  is in  $\mathcal{C}$ . If  $\mathcal{C}$  does not converge to any point,

then  $X$  is covered by  $N_p, p \in X$ , and hence by a finite number  $N_1, \dots, N_n$  of such sets. Therefore  $\phi = N'_1 \cap \dots \cap N'_n = (N_1 \cup \dots \cup N_n)' = X'$ , being the intersection of a finite number of sets in  $\mathcal{C}$  is also in  $\mathcal{C}$ . But this violates the definition of a filter.

Suppose that every ultrafilter of subsets of  $X$  converges. Let  $\mathcal{F}$  be a family of closed subsets of  $X$  with the finite intersection property. Let  $\mathcal{C}_1$  be the class of all sets which contain a finite intersection  $F_1 \cap \dots \cap F_n$  of sets  $F_i \in \mathcal{F}$ . It is evident that  $\mathcal{C}_1$  is a filter. Let  $\mathcal{C}$  be an ultrafilter refining  $\mathcal{C}_1$ , and let  $p$  be a point to which  $\mathcal{C}$  converges. Then, every neighborhood of  $p$  has a non-void intersection with every set of  $\mathcal{C}$ . Hence,  $p$  is in the closure of every set of  $\mathcal{C}$ . In particular,  $p \in F$  if  $F \in \mathcal{F}$ . Hence  $p \in \bigcap \mathcal{F}$ , so that  $\bigcap \mathcal{F} \neq \phi$ . The desired result follows from Lemma 5.6. Q.E.D.

### 8. Product Spaces

Given a Cartesian product (cf. Definition 3.11)  $X = \prod_{\alpha} X_{\alpha}$  of topological spaces  $X_{\alpha}$ , it is natural to seek to define a topology in  $X$ . For example, if  $X_1$  and  $X_2$  both represent the space of real numbers, the Cartesian plane  $X_1 \times X_2$  may be topologized by using sets of the form  $U \times V$ , where  $U$  and  $V$  are open, as a base. With this topology,  $X_1 \times X_2$  is two-dimensional Euclidean space. Abstracting a vital property of this example, we seek a topology  $\tau$  for the general Cartesian product  $X$ , such that each projection  $pr_{\alpha} : X \rightarrow X_{\alpha}$  is continuous. It is readily seen that the maps  $pr_{\alpha}$  will be continuous if and only if each set of the form  $U = \prod_{\alpha} U_{\alpha}$ , where  $U_{\alpha} \subseteq X_{\alpha}$  is open, and where  $U_{\alpha} = X_{\alpha}$  except for some single  $\alpha = \alpha_0$ , is in  $\tau$ . This property is not sufficient to characterize  $\tau$  completely. Among the various topologies which make the maps  $pr_{\alpha}$  continuous, we still have the three topologies determined by taking as a base the sets  $U = \prod_{\alpha} U_{\alpha}$  with:

(a) all  $U_{\alpha}$  open,

or

(b) all  $U_{\alpha}$  open,  $U_{\alpha} = X_{\alpha}$  for all but a countable set of  $\alpha$ ,

or

(c) all  $U_{\alpha}$  open,  $U_{\alpha} = X_{\alpha}$  for all but a finite set of  $\alpha$ .

With a view to future applications, the topology (c), which is the

weakest topology for  $X$  which will guarantee the continuity of the maps  $pr_\alpha$ , is selected for further analysis. The formal definition follows:

1 **DEFINITION.** Let  $(X_\alpha, \tau_\alpha)$  be an indexed family of topological spaces. Then the *product topology* is the topology  $\tau$  in  $X = \prod_\alpha X_\alpha$  obtained by taking the collection of all sets  $U = \prod_\alpha U_\alpha$ , where each  $U_\alpha$  is open, and where  $U_\alpha = X_\alpha$  except for a finite set of indices  $\alpha$ , as a base. A *product space* is a Cartesian product of topological spaces with its product topology.

It is apparent that a generalized sequence  $f$  in the product space  $X$  converges if and only if the generalized sequence  $pr_\alpha f$  converges in  $X_\alpha$  for each  $\alpha$ .

2 **LEMMA.** A product of Hausdorff spaces is a Hausdorff space.

3 **LEMMA.** If  $b \subseteq a$ , then the projection  $pr_b$  (cf. 3.14) of  $\prod_{\alpha \in a} X_\alpha$  onto  $\prod_{\alpha \in b} X_\alpha$  is continuous, and maps open sets into open sets.

4 **LEMMA.** A denumerable product of metric spaces  $X_n$ ,  $n = 1, 2, \dots$ , is a metric space. If  $\varrho_n$  is a metric for  $X_n$ , then a metric for the product space  $X = \prod_n X_n$  is given by the formula

$$\varrho(x, y) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{\varrho_n(pr_n x, pr_n y)}{1 + \varrho_n(pr_n x, pr_n y)}.$$

If each of the spaces  $X_n$  is complete, then the product space  $X$  with this metric is also complete.

The proofs of these lemmas are straightforward and elementary. They will be left to the reader.

5 **THEOREM.** (Tychonoff) A Cartesian product of compact spaces is compact in its product topology.

**PROOF.** The proof is based upon Lemma 7.12. Let  $X_\alpha$  be compact, and let  $\mathcal{E}$  be an ultrafilter of subsets of  $X = \prod_\alpha X_\alpha$ . Then the set  $\mathcal{E}_\alpha$  of all sets of the form  $pr_\alpha(E)$  with  $E \in \mathcal{E}$  is an ultrafilter of sets in  $X_\alpha$ . This is clear, since a proper refinement  $\mathcal{E}'_\alpha$  of  $\mathcal{E}_\alpha$  determines a proper refinement

$$\mathcal{E}' = \{E | E = pr_\alpha^{-1}(E_\alpha), E_\alpha \in \mathcal{E}'_\alpha\}$$

of  $\mathcal{E}$ . Lemma 7.12 shows that each  $\mathcal{E}_\alpha$  converges to some point  $x(\alpha) \in X_\alpha$ . It follows readily from the definition of the product topology that  $\mathcal{E}$  converges to the point  $x = \prod_\alpha x(\alpha)$  in  $X$ . Q.E.D.

## 9. Exercises

1 Any non-void open interval of reals is homeomorphic to the set of all real numbers, but not homeomorphic to a half open or a closed interval. A closed interval is not homeomorphic to the set of complex numbers of absolute value one.

2 Let  $R$  be the space of real numbers, and let  $R_x = R$  for each  $x \in R$ . Show that  $Q = \prod_{x \in R} R_x$  contains a countable subset whose closure is  $Q$ , but that  $Q$  does not have a countable base.

3 Not every normal space is a metric space.

4 Let  $G_1$  and  $G_2$  be open sets in a normal space, and let  $F$  be a closed set with  $F \subseteq G_1 \cup G_2$ . Show that  $F$  is the union of closed sets  $F_1, F_2$  with  $F_1 \subseteq G_1$  and  $F_2 \subseteq G_2$ .

5 Let  $S$  be an abstract set. Show that the collection  $B(S)$  of all bounded real or complex functions on  $S$  is a complete metric space under the metric function  $\varrho$  defined by  $\varrho(f, g) = \sup_{s \in S} |f(s) - g(s)|$ . If  $S$  is a topological space, then the collection  $C(S)$  of all bounded continuous real or complex functions on  $S$  is a closed subset of  $B(S)$ .

6 Establish the existence of continuous real functions of a real variable, such that

$$\limsup_{h \rightarrow 0+} \left| \frac{f(t+h) - f(t)}{h} \right| = \infty \text{ for all } t.$$

Hint: Let  $S$  be the real line and let  $C(S)$  be the space of periodic real continuous functions metrized as in Exercise 5. For  $m = 1, 2, \dots$  let  $C_m$  be the set of all  $f$  in  $C(S)$  for which

$$\left| \frac{f(t+h) - f(t)}{h} \right| \leq m$$

for some  $t$  and all  $h > 0$ . Using the Baire category theorem, show that  $C(S)$  is not the union of the sets  $C_m$ .